Network Interdiction — Work Log

Maintained by Claude Code and Codex for Anurag Ramesh

Last updated: 26 October 2026

This log records every change to the code, newest first. A new entry is added, and this PDF rebuilt, each time a piece of code is changed or implemented. Page 1 is always the current state.

Current state (25 October 2026)

Pull requests	
PR #20–PR #24	Merged: PR #20 and PR #21 on 6 October; PR #22, PR #23 and PR #24 on 15 October (by bernalde).
PR #25–PR #37	*Open.* All opened by bernalde on 15 October; CI green; no independent review on GitHub. Reviewed locally on 22 October (entry below): none is ready as is. PR #29, PR #33 and PR #37 need changes; the others need small fixes. Nothing has been posted to GitHub.
Merge order	Two stacks: PR #25→PR #27→PR #29→PR #36→PR #37 and PR #28→PR #30→PR #31→PR #32→PR #35. PR #26, PR #33 and PR #34 stand alone. PR #37 already contains PR #28 and PR #30–PR #35, so it must not be merged first.
Gurobi	*Fixed* on 22 October: academic license 2877481 (expires 2027-10-22), bound to the fixed-address adapter `eth1`.
Results notebook	*Rebuilt* on 23 October from merged results only, on the local branch `notebook/results-v2` (not pushed). On 25 October, added expected best objective gap against measurement count for the tiny QAOA states; existing figures are unchanged. Sections for open PRs are added as each merges.
Long runs	None active.
Decomposition benchmark	The eight-thread run (14 October, 497 rows, all 27 slide cases) is checked and committed on `notebook/results-v2` as `results/decomposition_large.csv`.
Backlog (only on request)	
1	Follow-up fix for the merged PR #24: add the empty-cut probability and a warm-start-only baseline, report $P(\text{opt})$ first, and disclose or drop the λ feature.
2	Post the review findings on each open PR (needs your approval).
3	Cap `partition_to_fit` when no partition fits.
4	LP line, led by the user (decided 23 October): take over PR #25's branch, then study why the LP is exact on these instances, where it breaks, and LP-seeded QAOA.
5	Deferred: implementing arXiv 2610.10772. (Benders, #15, is now in PR #27.)
Needs you	
	Review the rebuilt notebook; then push `notebook/results-v2` and open a PR, or ask for changes.
	PR #26: confirm the MIT license with Hanjing Xu, who wrote most of the original code.
	Drop the two old stashes (`git stash drop`, twice); I was not allowed to.
	Decide whether to commit the IBM hardware results and this log to the repository.

Entries

Copy the project to SECQUOIA

26 Oct 2026

Request Move the repository to SECQUOIA, or create a new repository there and copy the project if a transfer is not possible.

Change GitHub rejected the direct transfer with HTTP 422 ("Repository can't be transferred"). Created the private repository SECQUOIA/Network-Interdiction. The copy includes the current notebook, results, paper, V1 presentation and worklog on `notebook/results-v2`, alongside the source repository's existing branches and Git history. The current-work branch is the new default; the source main branch is preserved. Local environments, caches and ignored build files are excluded.

Scope Issues and pull requests remain in the original repository; their historical links in this log still point there. No pull requests were merged and no experiments were rerun.

Verification Confirmed that the destination is private and that all 169 project files in the prepared copy match the working checkout byte for byte. Preserved tracked file modes and rebuilt this PDF.

Keep only the V1 presentation

26 Oct 2026

Request Remove all other presentation versions, their corresponding PDFs and scripts, keeping the version ending in `_V1`.

Change Preserved the author's `Network_Interdiction_Project_Update_V1.pptx` unchanged and exported its matching three-page PDF with native PowerPoint. Removed superseded decks, the old `st_NIP` presentations, PDFs, previews, figure copies, archived checkpoints, generation metadata and version-specific scripts. The slide README now points only to V1. Font files and their license remain. The notebook, source figures, paper and research code are unchanged.

Verification Checked the V1 package, confirmed the PDF's page count and titles, and verified that the retained PPTX has the same SHA-256 hash as before cleanup. V1 is the only remaining project presentation.

Use the exact notebook plots in a three-slide presentation

26 Oct 2026

Request Replace the separately redrawn slide figures with the same plots as the notebook. The user selected three slides when asked about readability and the earlier two-slide limit.

Change Captured the latest author-edited deck. Slide 1 pairs the size and sampling figures; slide 2 shows decomposition; slide 3 shows parameter setting strategies at both $p = 1$ and $p = 2$. Each embedded PNG is byte-for-byte identical to its saved notebook figure. No plot is redrawn, cropped or relabeled. The latest author titles, wording, formatting and removal of the visible MPS sentence are preserved across the new layout. The transfer caption and notes clarify which depth its 66% statistic describes.

Files The current presentation, PDF and previews use suffix `_notebook_plots` in the existing slide directory. The new builder copies saved figures directly; its manifest records their hashes. Earlier presentations and all notebook files are untouched.

Verification Verified all four image payloads against the notebook PNGs, checked aspect ratios and zero crop, and checked preservation of author text and unrelated package parts. Native PowerPoint exported three slides and three PDF pages. All previews were inspected; package, template and delivery checks pass, with no native text overflow.

Add the qubit-size motivation and refresh the presentation plots

26 Oct 2026

Request Include the notebook's problem-size figure as a main reason for decomposition, and update the presentation with the cleaned plot labels.

Change Started from a new checkpoint of the author's latest saved deck, including the title "Decomposition-Based Refinement". The first slide now places the full-QUBO and fitted-piece sizes beside the two decomposition comparisons. It retains all network points, family markers, the device and simulation limits, and the no-fit warning. Both QAOA plots are refreshed with concise axis labels, and the SEM bars remain. The presentation still has two slides. All existing wording, formatting and notes are preserved; only figure layout and two result-text positions change. Added figure provenance to the first slide's notes.

Files The outputs are in `slides/network_interdiction_update/`, with suffix `_updated`. Earlier author-edited presentations are untouched. The revision builder, input hashes, checkpoint and preservation report are stored alongside them.

Verification Checked that unrelated OOXML parts and existing text remain unchanged. Rendered both slides in native PowerPoint, inspected the plots and labels, and ran package, template and PDF-delivery checks. No notebook data or experiments changed.

Concise axis labels throughout the results notebook

26 Oct 2026

Request Apply the axis-label cleanup to all notebook plots, without secondary explanations in the labels.

Change Removed the QUBO size formula, mean-count parentheticals, log/symlog notes and decomposition-size ranges from the axes. Kept the primary quantities, units, and the distinction between the two decomposition settings. The stacked outcome plots now use "Probability". Refreshed the five affected saved figures and their embedded notebook PNGs. Data, scales, tick labels, legends, error bars and notebook prose are unchanged.

Verification Checked that every other notebook field and saved figure remains unchanged, preserving the user's edits. Ran the existing notebook checks against the source data.

Shorter y-axis label on the sampling-quality plot

26 Oct 2026

Request Remove the mean/SEM parenthetical from the y-axis title.

Change The label is now "Expected best gap to optimum, %". The SEM bars, data, legend and notebook explanation are unchanged. Refreshed `figures/qaoa_sample_quality.png` and only that plot's embedded PNG in the current notebook, preserving the user's other edits.

Verification Checked that all other notebook content and saved figures remain unchanged. The existing sample-quality check verifies the mean curves and SEM endpoints against the six per-case values.

Taller decomposition plot, preserving the author's slide edits

26 Oct 2026

Request Explain slide 1's categorical budget-allocation axis and fix its overly wide, shallow plot while preserving the user's changes.

Change Captured the latest saved presentation; it was not open in PowerPoint. Rebuilt only the decomposition image with more vertical space, kept its aspect ratio, and moved the two introductory lines upward into space beneath the user's shorter title. The revised PPTX and PDF have the suffix `_plot_fixed` in `slides/network_interdiction_update/`. The user's original file is untouched. Titles, removed text, formatting, notes and all of slide 2 are preserved. No notebook or result data changed.

Verification Package comparison limits changes to slide 1's plot image and three shape geometries. All other package parts, including slide 2 and notes, are byte-for-byte unchanged. Checked package integrity and rendered the revision in native PowerPoint.

Two-slide SECQUOIA project update

25 Oct 2026

Request Create at most two slides using the SECQUOIA deck, explaining the supplied project progress report and the notebook's experimental results.

Content Slide 1 covers the interdiction objective, penalty validation, partitioning, budget allocation and refinement. It reuses the decomposition figure to show the trade-off between small pieces and solution quality, alongside the exact and greedy classical references. Slide 2 pairs the expected best-of-N gap figure (including SEM) with the depth-two transfer outcome panel, and summarizes MPS and the separate IBM Fez pilot. The warm-start claim is qualified: lower QUBO energy did not imply better interdiction, and much of the probability remained on the empty plan. Speaker notes explain the cohorts, metric, invalid-plan handling, simulation limits and hardware provenance.

Files `slides/network_interdiction_update/` contains the editable presentation, matching PDF, slide previews, builder, native PowerPoint export script, plot copies, and a manifest of input hashes. The builder uses the neighboring SECQUOIA template v5.19. Roboto fonts are embedded in the delivered presentation; the font license is included. The original notebook, figures and experimental data are unchanged, and no experiments or hardware jobs were run.

Verification Native PowerPoint exported exactly two slides and two PDF pages. Package integrity, SECQUOIA publication and delivery checks pass. Both slide previews were inspected; all 21 native text shapes fit their boxes within a one-point tolerance. Delivery and layout reports are saved under `qa/`. Both PowerPoint files retain speaker notes, and the PDF retains figure descriptions; these checks are not an accessibility certification.

SEM bars on the QAOA measurement-count figure

25 Oct 2026

Request Add standard-error bars to the new solution-quality figure.

Change Each point now shows its mean ± one SEM across the six cases, using the sample standard deviation (ddof = 1) divided by $\sqrt{6}$. These bars describe variation across cases, not shot noise. The axis label, notebook explanation and README state this definition. The saved data and mean curves are unchanged.

Verification The notebook test independently computes SEM from the six per-case values and checks both endpoints of every plotted error bar. All 22 notebook checks pass.

QAOA solution quality against measurement count

25 Oct 2026

Request Implement the final suggestion in the user's pasted Claude conversation: add a figure of the best-of-N objective gap against the number of measurements, with one line per QAOA depth and uniform guessing as the baseline. Keep the existing outcome figure.

Method `qaoa_sample_quality.py` re-simulates the saved tiny-instance angles, marginalizes auxiliary bits, and scores each interdiction plan by $f/f^* - 1$. An over-budget plan receives the empty plan's score, maxflow$/f^* - 1$; it cannot earn credit by cutting extra arcs. An exact survival-tail calculation gives the expected minimum gap after N independent measurements. Each curve averages those expectations over the same six cases, with equal weight per case. No angles are retrained, and no greedy repair is applied. This measures ideal sampling quality, without training time, execution time, or hardware noise.

Data and notebook `results/qaoa_sample_quality.csv` has 288 rows: six cases, uniform guessing and depths one through three, and twelve measurement counts from 1 to 4096. Its header records the working-tree commit, framework versions, and hashes of the inputs and scoring code. The notebook only reads this CSV. The new figure is `figures/qaoa_sample_quality.png`; README explains its metric and regeneration command. The existing outcome plot and every older figure are byte-for-byte unchanged. Changes remain local on `notebook/results-v2`; nothing is pushed.

Findings At ten measurements, mean expected gap is 49% for uniform guessing, 19% at $p = 1$, 9% at $p = 2$, and 10% at $p = 3$, reproducing Claude's scratch calculation. More measurements help; more depth is not consistently better. With thousands of measurements the gaps approach zero on these tiny cases, concealing their different sample requirements. These expectations do not guarantee the result of a particular shot batch.

Verification All 48 notebook, outcome, and sample-quality tests pass. Checks include direct enumeration of measurement sequences, independent NetworkX flow calculations, a Qiskit Statevector cross-check, coverage and monotonicity of the saved curves, and the plotted means. Three in-memory defects—rewarding an illegal cut, ignoring N, and reversing the x-bit order—each fail the intended test. The notebook executes without errors, the new figure was inspected, all eleven input and source hashes match, and all 21 existing figure hashes are unchanged.

QAOA depth axis label spelled out

24 Oct 2026

Change At the user's request, the left panel's y-axis now reads "Probability of optimal plan" instead of "P(optimal plan)"; nothing else changed (commit `68b3286`). The notebook executes without errors and its 21 tests pass.

Figure titles removed

23 Oct 2026

Request The user asked for no titles on any of the plots.

Change No figure has a title or suptitle any more. Panel identifiers moved to axis labels (QAOA depth p, decomposition setting) or a caption under each example network. Each takeaway the titles stated is now a sentence under its figure, and a figure-size takeaway was added for the problem-size plot. README and the module docstring say so (commit `e4f2f61`).

Verification The notebook executes without errors and 21 tests pass. Two are new: one checks that no figure has a title, and one recomputes every number the text quotes (19 claims) from the raw CSVs; a wrong number planted in the text made it fail.

Gurobi timed at its first optimal incumbent

23 Oct 2026

Request The user asked whether the time-against-gap figure used Gurobi's best-incumbent time from the solver callback, rather than the time to prove optimality. It did not: it used `Gurobi_time_sec`, the end of the solve.

Change Gurobi is now placed where its MIPSOL callback first reported the optimal incumbent (`Gurobi_trajectory`). That matches the heuristics, which are timed until their answer is ready. Across all 27 cases the best incumbent is the optimum, found at 0.0085–0.048 s; the proof follows within 0.7 ms, so the points barely move. The title, the axis label and the notebook text say what the times mean (commit `68223eb`).

Verification The notebook executes without errors and its 19 tests pass. The test recomputes the incumbent times from the trajectory JSON; with the proof time planted back, it fails.

Notebook reloads its figure code

23 Oct 2026

Problem The user still saw the unlabelled figure. The notebook on disk was correct, but Run All in a kernel that had already imported `results_plots` reuses the old module, so figure edits did not show without a kernel restart.

Change The setup cell reloads `utils` and `results_plots`, as the earlier notebook did for `utils` (commit `f28028d`). The notebook executes without errors, and its 19 tests pass.

Large networks labelled in the problem-size figure

23 Oct 2026

Request After the problem-size figure was explained, the user asked for each large network to be labelled. Three of them (planar 125, ER 90 and k-NN 100) nearly coincide, and the 50-node ones cluster.

Change The nine names now sit in a column to the left of the group, in the same top-to-bottom order as the points, each with a leader line. The planar 125 label carries the "no split fits 155 qubits" note. Commit `9fbdadf` on `notebook/results-v2`.

Verification The notebook executes without errors. 19 notebook tests pass, including a new check that each network is named once, at its own point.

Notebook names the instances that were run

23 Oct 2026

Request The user asked the notebook not to refer to slides or slide instances, and to say which instances were actually run.

Change The notebook opens with a table of the four instance sets it uses: tiny (4 networks, 6 cases), the transfer cohort (6 graphs), mid-size (9 networks, 23 cases) and large. The large set is the nine networks of `testcases/`: planar with 50, 75 and 125 nodes, Erdős–Rényi with 50, 70 and 90, and k-NN with 50, 75 and 100, with 288–766 arcs and budgets 4–12 (27 cases). That name replaces "slide" in the text, the figure titles and legends, and the code (`load_large`). `results/decomposition_slide.csv` became `results/decomposition_large.csv`, with its data bytes unchanged (commit `76df400`).

Verification The notebook executes without errors and contains no mention of slides. The 34 notebook, outcome and decomposition tests pass. The two retitled figures were inspected.

Results notebook rebuilt from merged results

23 Oct 2026

Request The user found the notebook's plots unhelpful and asked for a proper results notebook, built first from merged PRs only, with each open PR added after it merges.

Branch Local branch `notebook/results-v2` from `main`, not pushed. `47e1e63` keeps the uncommitted 14 October notebook work unchanged as the starting point. `b012a4e` adds `qaoa_outcomes.py`, `6286beb` its results, `adc49c4` the decomposition results, and `9478133` the notebook, `results_plots.py`, tests, README and figures.

New data `results/qaoa_outcomes.csv` (96 rows) re-simulates every stored QAOA state exactly and splits it over interdiction plans: optimal, within budget, empty, or over budget. All 84 stored P(optimal) values reproduce within 10^{-6}; it was generated from a clean checkout. `results/decomposition_slide.csv` is the finished 14 October eight-thread run (497 rows). Its data bytes are unchanged, and it is checked against the slide optima, greedy values and max flows.

Notebook Eight figures, each with a title computed from its data: example networks with their optimal cuts; QUBO size against simulator and device limits; a heat map of heuristic gaps; time against gap; QAOA's outcome split by depth; the parameter-transfer outcome split; within-budget and optimal MPS shots; and decomposition gaps by budget rule. The notebook runs in about 15 s and calls no solver. New figure names avoid overwriting the older figures, which stay in `figures/` unused.

Findings Gurobi proves all 27 slide optima within 0.05 s. Greedy is optimal on 19 of 27; SA + repair never beats greedy. On the tiny instances, QAOA's P(optimal) rises from 0.06 to 0.22 by $p = 3$, but the empty plan grows from 0.02 to 0.38, so energy scores mislead. On the held-out cohort, the transfer methods match target training, and RWS puts 66% of its probability on the empty plan at $p = 2$. Above 100 qubits, at most 0.4% of MPS shots stay within budget. Four pieces with refinement reach the optimum in 21 of 27 cases; 155-qubit pieces leave a median 22% gap even with the best rule (min-cut).

Verification Notebook tests: 19 pass, checking every figure against the raw CSVs; four planted bugs in the figure code were each caught. Outcome tests: 8 pass, and two planted bugs were caught. Decomposition tests: 7 pass. Full suite with the real license: 264 passed, 9 skipped (optional torch and `qaoa_training_pipeline`).

Independent review of PR #24–PR #37

22 Oct 2026

Request The user asked not to trust the PRs and to review each one carefully. No repository code was changed.

Method Nine read-only reviews, each in its own temporary worktree (all removed afterwards). Each read the PR's diff against its own base, checked its claims against the linked issue, re-ran its tests and the full suite with Gurobi's trial license, recomputed committed numbers, and planted bugs to check that the tests catch them. Nothing was posted or changed on GitHub. The main checkout was not modified: one stray command in it was checked and changed nothing.

Verdicts

PR	Verdict	Main problem
PR #24 (merged)	follow-up fix	Misleading RWS score; transfer no better than one fixed angle set; the λ feature uses the exact solution
PR #25	small fixes	The Lagrangian pass has no time limit
PR #26	small fixes	Needs the consent of Hanjing Xu, author of most original code
PR #27	small fixes	Stopping test untested; ignores the thread flag; unconverged runs plotted as converged
PR #28	small fixes	D-Wave occurrence counts are always 1 after descent
PR #29	needs changes	The shared price is never applied to the piece solves
PR #30	small fixes	`QCI_API_URL` is required, not optional; failures lose finished rows
PR #31	small fixes	QAOA numbers depend on the machine; one large case aborts the batch
PR #32	small fixes	No guard against QPU spend; the 300 s timeout loses paid results
PR #33	needs changes	Large-case embeddings depend on machine speed; 884 KB of regenerable data
PR #34	small fixes	The Lean proves the arithmetic steps only; docs claim the whole bound
PR #35	small fixes	XY is no better than a random size-B cut; the table omits that baseline
PR #36	small fixes	"Interior" arcs are not interior; the one hard case is formulation-dependent; data licenses
PR #37	needs changes	Contains seven other PRs; SA understated; the large QAOA run was never trained

Main finding The current instances are easy for classical methods. The LP relaxation equals the optimum on 49 of the 50 slide and mid cases, and LP rounding plus repair is optimal on all 50 in about 0.02 s each. PR #36's one "hard" case is solved in 0.1 s by a simple exact search and in about 3.5 s by the standard MILP. The "hard for Gurobi" goal of #1 is not met yet.

Correction The RWS warm-start score reported for PR #24 (gap closed 0.80 and 0.86) mostly reflects probability on the empty cut. That cut leaves the full flow, the worst answer, yet scores about 0.98. On $P(\text{opt})$, RWS is worse than standard QAOA at $p = 2$ (0.118 against 0.138). GAT and IBM transfer were no better than one fixed angle set.

Tests Every PR's full suite passes with zero failures. The remaining skips need optional packages (torch, IBM, QCI) or, at the time, the full Gurobi license.

Side effects The PR #34 review installed the Lean 4.35 toolchain in `~/.elan` and filled the mathlib cache.

Gurobi license fixed permanently

22 Oct 2026

Cause On WSL2, Gurobi's host ID is the last four bytes of `eth0`'s hardware (MAC) address, and WSL2 gives `eth0` a new random address at every start. Each key therefore stopped working after the next restart (HostID mismatch on 16 and 20 October).

Fix Following Gurobi's WSL2 recipe, a boot service (`gurobi-fixed-adapter.service`, set up by the user with `sudo`) creates an unused adapter `eth1` with the fixed address `02:00:47:52:42:01`. On the first attempt, udev's default address policy overwrote that address; the rule `/etc/systemd/network/10-gurobi-eth1.link` now prevents this, and the script waits for udev before setting the address. The adapter has no IP address and does not affect networking.

License Installed Gurobi's license tools 13.0.3 in `~/.local/bin`. Activated the new key against `eth1` (host ID `47524201`): license 2877481, academic, expires 2027-10-22. The old file is kept as `~/gurobi.lic.bak-20261022`.

Verification A 3000-variable model, above the trial license's 2000-variable limit, solves. The two tests that need a full license, `test_lambda_window` and `test_slide_results`, now run and pass (28 passed). Survival across a WSL restart was not tested, at the user's choice. If the license breaks again, check `ip -o link show eth1`; if the address differs, restart the service. No repository code changed.

PR #22–PR #24 merged; thirteen new PRs (recorded 22 Oct)

15 Oct 2026

Merges bernalde merged PR #22, PR #23 and PR #24 into `main` on 15 October (head `56e66d6`). PR #24 had no independent approving review; it was reviewed afterwards (entry above).

New PRs bernalde opened PR #25–PR #37 the same day. They cover LP, Lagrangian and Benders baselines (#15), the shared budget price (#14), real networks (#1), QCI, Qiskit and IBM backends (#17), embedding cost (#4), penalty bounds with Lean (#2), the XY mixer (#3), raw annealer feasibility (#18) and the license (#9).

Decomposition The eight-thread decomposition run started on 14 October finished at 21:33 that day: 497 rows covering all 27 slide cases. It has not been verified or committed.

Notebook metrics aligned with quantum preconditioning

14 Oct 2026

Scope The user requested a simpler notebook using the same metrics and plots as the quantum-preconditioning repository. Further experiments and implementation work require a new user request.

Definitions Read `Gurobi_QP/notebooks/analysis.ipynb` and its `qp_gurobi/utils.py`. The notebook now measures the original feasible interdiction objective, surviving maximum flow f, against its exact optimum f^*. It uses $f/f^* - 1$ and $100[1 - (f - f^*)/|f^*|]$, matching the reference notebook's executable definitions. Zero optima are excluded. These are distinct from the earlier energy scores and normalized flow-reduction plots.

Plots Reduced the main notebook to three figures: the five-method objective gap at depths one and two after repair, the classical performance distribution, and the recorded incumbent trajectory. All saved small transfer results tie at the optimum after repair. Timing/PAR and scaling comparisons that need missing trajectories are not reconstructed from final scores. Existing diagnostic helper regressions remain covered separately. This change reads saved data and does not run experiments.

Verification All ten notebook and metric checks pass. They retain the earlier helper regressions and check the new objective formulas, missing/zero references, independence from Hamiltonian scores, plotted trajectory values, and every suboptimal performance point. The notebook was executed from saved CSVs and its three rendered figures inspected.

Depth-one and depth-two comparison for presentation

14 Oct 2026

Request Added a side-by-side comparison of all five QAOA methods at $p = 1$ and $p = 2$, using the same six held-out graphs and the saved 60-row statevector experiment. Depths occupy columns; energy gap closed and optimal-interdiction probability occupy rows. Axis scales match across depths, graph colors match across panels, and black diamonds mark means.

Artifacts Updated `paper/make_results.py` to produce a standalone depth-one figure and a 16:9 comparison in PNG, PDF, and SVG formats under `paper/figures/`, named `transfer_depth_comparison.*`. The comparison PNG uses 300 dpi. The manuscript retains its depth-two figure. No experiment was rerun, and no saved measurements were changed.

Eight-thread Gurobi policy and full benchmark restart

14 Oct 2026

Request The user requested eight threads for every Gurobi run and then explicitly requested a complete restart for fair timing comparisons. Stopped the previous decomposition process (PID 108041). Its 153 rows covering 16 case/setting groups, log, command and checksum are archived in `logs/decomposition-before-eight-threads-20261014-145615/`. No old timing row will be resumed or combined with the fresh run.

Configuration A shared model factory explicitly sets `Threads=8` for exact NIP references, penalty checks, Lagrangian solves, restricted piece models, and the trajectory solver. The benchmark CLI now defaults to eight instead of one. Its existing explicit override applies to the dedicated trajectory solve; reference models remain at eight. The decomposition CSV records its thread budget, refuses to overwrite an existing output, and logs partition/contraction versus piece-solve phases. The notebook identifies its historical one-thread timing data separately.

Restart The fresh run starts at the first of all 27 cases, with both partition settings and budget refinement. It runs from the clean checkout `/tmp/ni-decomp-eight`, source `1ce5438`, as PID 169748, starting at 15:01 EDT. Output, log, and launch metadata are in `logs/decomposition-eight-threads-20261014-150145/`. Verified `Threads=8` on a constructed model before launch; the CSV records that setting and the source hash. The first two cases produced 20 new rows. No partial row from the prior run was resumed. The observed initial process CPU usage exceeded two cores during the piece-solve phase.

CPU interpretation The previous decomposition models used Gurobi's automatic thread setting. Observed one-core CPU use was not a confirmed one-thread Gurobi limit. NetworkX partitioning, exact terminal contraction, and max-flow calculations remain serial; eight Gurobi threads do not parallelize those stages.

Verification The new runtime tests inspect each real model at `optimize()`, including the CLI and nested reference solves. All seven new checks pass. A scratch mutation forcing one thread fails all six solver-path checks, while the overwrite-protection test still passes. All 59 relevant solver, trajectory, decomposition, and penalty-window tests pass. Both PR #23 CI jobs pass; all PR #24 checks also pass, including both workflow runs triggered by the base/head updates. Posted and read back the restart summary on PR #23. The paper's decomposition protocol now records the eight-thread setting.

SIAM manuscript and rendered PDF; transfer results complete

14 Oct 2026

Paper Drafted `paper/Main.tex` with Anurag Ramesh and David E. Bernal Neira as the user-confirmed authors. It uses the neighboring quantum preconditioning project's unchanged SIAM 2017 class and bibliography style. A separate compatibility file restores standard LaTeX reference hooks for current hyperref/cleveref. The rendered PDF and Overleaf ZIP are complete; the PDF has eleven pages, with penalty and contraction proofs, implementation-specific methods, two result tables, a two-panel figure, limitations, and primary references. The writing follows the requested approximate ASD-STE100 style while retaining mathematical terminology. The PDF is a working draft. Verified the rendered title, proofs, tables, figure and references; there are no unresolved references or overflowing text. The Overleaf archive also compiles independently after extraction.

Data The transfer experiment finished all 60 rows from clean source `ac571aa`: 12 training graphs, six held-out graphs, depths one and two, five methods. Its largest statevector has 21 qubits (2^{21} complex amplitudes, 32 MiB at double precision before simulator overhead). All 28 transfer and saved-artifact checks pass, including recreated energies, donor choices and a saved GAT on a contracted piece. Four artifact mutation probes (missing row, wrong energy gap, wrong energy, and zeroed GAT weights) each fail the intended checks; the untouched scratch baseline passes all seven artifact tests.

Results At depth two, mean gap closure is 0.6528 (fixed), 0.6511 (IBM), 0.6621 (GAT), 0.6695 (target-trained), and 0.8610 (warm start). Warm-start optimal-interdiction probability is lower than target-trained QAOA (0.1183 versus 0.1377). All repaired outputs are optimal; greedy is exact because the cohort uses budget one. These results do not demonstrate an advantage on hard interdiction. The paper uses the common ten-instance subset for the MPS train/donor comparison, avoiding mismatched cohorts.

GitHub Committed the raw CSV/JSON/checkpoints, seven artifact tests, CI command, and result interpretation as `d5152ef`; pushed it and the penalty precision fix `da352f5` to PR #24. Original CSV line endings are preserved; the whitespace check passes with CRLF recognized. Checked four implemented #16 tasks and verified the read-back. The first task ("land" QAOA) stays open until PR #22 merges. All three CI jobs pass at this head. Posted and verified the result summary, then marked PR #24 ready and read back its state. No PR was merged.

Pending The original classical decomposition process remains active. The paper describes its protocol and labels the performance results as pending; no partial decomposition CSV is used in the manuscript. Worklog, slides, hardware logs, and the new paper remain local, outside the transfer PR. The manuscript still needs the full decomposition comparison, harder instances, multiple seeds, a product-state-only ablation, MPS convergence, and complete runtime accounting before submission.

Transfer PR opened; decomposition result checks prepared

14 Oct 2026

State PR #24 is open on top of PR #23. GitHub's core, QAOA and transfer jobs all passed at `ac571aa`. The 12-training/6-test experiment at depths 1 and 2 is running from `/tmp/ni-transfer-benchmark`, with outputs in `/tmp/ni-transfer-output` and log `/tmp/ni-transfer-benchmark.log`. Its provenance records a clean commit.

Decomposition The original benchmark is still running. Its first 133 rows cover 14 of 54 case/setting combinations (the 50- and 75-node planar cases). A separate PR #23 checkout at `/tmp/ni-finish-pr23` holds draft result checks for complete case coverage, ratios, budgets, qubit limits and monotonic refinement. A copy of the partial table fails the coverage check, as intended; it is not committed. The active run is untouched. A new status comment on PR #23 records the verified handoff.

Result checks Prepared transfer-table checks for all 60 expected rows, disjoint graph splits, donor-angle provenance, zero target optimizer evaluations for transferred schedules, independently enumerated optimal flows, energy gaps, and reproduction of a held-out energy for each method. The comparison uses budget one, where greedy is exact; interpretation is limited to raw quantum energy/probability scores, not a claim of improved repaired interdictions.

Reference correction The restricted Gurobi curve now removes MIP tolerance noise for integer capacities before comparing penalty-window endpoints, while retaining fractional objectives for fractional-capacity callers. Added an integer/fractional reference comparison against exhaustive enumeration; all 21 transfer tests pass. Committed as `da352f5`. Prepared saved-model checks to reproduce the reported donor choice on each held-out graph. The saved GAT passed a real contracted-piece solve (flow 5 on `tiny_a`, with only owned arcs cut).

Parameter-transfer implementation and verified issue checklists

14 Oct 2026

What Checked nine completed tasks across #2, #4, #14, #15, and #18, with each updated issue read back from GitHub. The "every infeasible assignment is worse than every feasible one" task and the closed-form pipeline task remain unchecked: the exact two-sided λ window does not establish those stronger claims.

Implementation Added flow-network features (capacities, direction, source/sink, budget, penalty, fixed-arc ownership), the actual IBM `TransferTrainer` adapter, a three-layer GATv2 donor classifier adapted from the MIS code, and regularized warm starts with a matching tilted mixer. QAOA now supports restricted interdiction sets for contracted pieces. Committed as `ac571aa`, with optional dependencies and a dedicated transfer CI job. A clean detached checkout runs the result generation.

Experiment Added `run_parameter_transfer.py`: graph-level training/test separation, fixed-donor, IBM, GAT, per-instance COBYLA and RWS comparisons; CSV metrics, graph/angle provenance and JSON model checkpoints. A four-training/two-test, depth-one smoke run completed. The full comparison will run from a clean committed checkout.

Tests Before editing code, the current full suite ran with both QAOA backends required: 181 passed, 17 skipped in 155 seconds. Licensed tests skipped in the sandbox. New integrations use an isolated temporary Python environment, preserving the environment of the active benchmark. The initial 17 transfer tests pass. In a separate scratch copy, six deliberate defects were each caught: unrestricted repair, wrong mixer sign, wrong regularizer sign, missing budget feature, farthest-donor selection, and discarded GAT edge features. The working tree was not mutated. The full licensed suite passed all 214 tests in 217 seconds; the final transfer module passed 20 tests in 12 seconds, including the restricted Gurobi reference and saved-bank reload. The regular environment without optional transfer dependencies passed 11 tests and skipped five.

Sources Verified MIS commit `ab8434f`; retained its Apache license. The current IBM commit `4bad7bc` uses `provide_params` with depth set at construction, unlike the older local clone. The dependency is pinned. Regularized warm starts follow equations (9) and (18) of arXiv:2603.10191, adapted to this repository's normalized minimization Hamiltonian.

Handoff: benchmark fix and mid-size results are pushed

14 Oct 2026

What Claude committed `0dd3cde`: `run_decomposition_benchmark.py` now writes a "no fit" row and carries on when a partition cannot meet the qubit limit; `--fit-qubits` sets that limit. Commit `bc19e01` adds `results/qaoa_mid.csv`: 69 donor-angle rows and 30 directly trained rows. The local branch and GitHub PR #23 both point to that commit.

Tests The handoff records two passing benchmark tests and four passing result-consistency tests, with four deliberate CSV corruptions caught and the original bytes restored. Codex is verifying current checks and the remaining results before completing the PR.

State The restarted decomposition run is active in its existing detached worktree at `0dd3cde`; it is being preserved. The issue-body edits were prepared but never posted. Only verified completed tasks will be checked: exact-window computation replaces bisection but still needs exact solves, so it does not complete the request for instance-only closed-form bounds. This work log remains local, as before.

Work log started

10 Oct 2026

What This document: `worklog/worklog.tex`, built into `worklog/worklog.pdf` with `pdflatex`. Not committed to git.

PR #23 (draft): mid-size instances, MPS QAOA, better decomposition

8 Oct 2026

Why There were no instances between `tiny_c` (22 qubits) and `50.planarflow` (912), so nothing beyond the tiny ones could run on a simulator or on a 156-qubit IBM chip.

What

- `instances.py`: nine seeded networks in `testcases/mid/` (planar, Erdős–Rényi, k-NN), 39–149 qubits. Their exact optima for 23 instance–budget rows are in `results/mid_instances.csv`. Source and sink: the pair with the most arc-disjoint paths.
- QAOA past 24 qubits: matrix-product-state (MPS) simulation (`qaoa_qiskit.py --method mps --bond-dimension`), a qubit reordering that keeps couplings short (reverse Cuthill–McKee), exact references from Gurobi, scoring from shots, and a new score `gap_closed` (0 = random guessing, 1 = best state). `run_qaoa_mid.py` runs all 23 rows.
- Decomposition (`decomposition.py`): a `marginal` budget split (solve each piece first, give cuts where they save most), `iterated` solving, `refine_split()`, `partition_to_fit()` (fewest pieces that fit a qubit limit), an exact Gurobi solver per piece. `run_decomposition_benchmark.py` compares every rule with Gurobi's optimum.

Tests 193 pass locally; CI green (QAOA job 41 passed; main job 166 passed, 27 skipped). 16 deliberately reintroduced bugs were all caught.

Results so far

- **Mid-size instances are easy classically.** Greedy is optimal on all 23 rows, and on 22 the optimal cut touches the source or sink.
- **MPS speed.** The reordering is essential: one 78-qubit evaluation had not finished in 10 minutes without it; with it, 40 qubits take 0.1 s. At 116 qubits one evaluation takes 15 s (bond cap 32) or 92 s (cap 64).
- **QAOA at 39–149 qubits** (angles trained on `tiny_a`, reused unchanged):

depth	energy gap closed (median, range)	shots within budget (median)	optimum after repair
p = 1	0.40 (0.33–0.49)	0.000	18 / 23 rows
p = 2	0.54 (0.47–0.67)	0.004	18 / 23 rows
p = 3	0.63 (0.56–0.79)	0.053	18 / 23 rows

  Training the angles directly on the 30 rows where that was affordable (up to 81 qubits) closes only 0.031 more of the gap (median; range 0.006–0.368). So transferring angles costs little. The catch: almost no raw shot respects the budget (the linear λ penalty, #2); the optimum appears only after greedy repair.
- **Decomposition, partial** (each piece solved exactly):

instance	pieces	budgets	best variant optimal	marginal, iterated optimal
`50.planarflow`	4	4	4	2
`75.planarflow`	4	3	3	3
`125.planarflow`	4	2	2	2
`50.planarflow`	8 (fits 156 qubits)	4	3	2
`75.planarflow`	30 (fits 156 qubits)	3	2	2

  Earlier spot checks on k-NN and Erdős–Rényi instances were mixed: refinement found the optimum on `75_6.2dnnflow` (145) where greedy does not (165), but the `marginal` split was much worse than `mincut` on `70_0.1.erdosflow` (100 vs 37; optimum 32).

Open Benchmark crash on `125.planarflow` (see Current state); result files not yet committed.

IBM Quantum hardware run on `ibm_fez`

8 Oct 2026

What Verified access (Open Plan, account `open`). Ran the trained QAOA circuits on the real device, 4096 shots each. Cost 8 s of QPU time (jobs `db3pmc2mb58s73886ir0`, 3 s; `db3pmmo4qg6s73c15qc0`, 5 s), logged in `logs/ibm_jobs.jsonl`. Nothing committed.

circuit	2q gates	P(optimal): hw / ideal	random	approx. ratio: hw / ideal
`tiny_a` B=1, p=1 (14 q)	41	0.257 / 0.260	0.125	0.789 / 0.807
`tiny_a` B=1, p=2	70	0.311 / 0.333	0.125	0.869 / 0.894
`tiny_a` B=1, p=3	108	0.355 / 0.322	0.125	0.896 / 0.930
`tiny_d` B=1, p=1 (21 q)	66	0.091 / 0.101	0.031	0.778 / 0.799

Finding Finding the optimal cut survives the noise (2–3× random guessing, close to ideal); the energy degrades a little with depth.

CUDA-Q at 21–22 qubits

7–8 Oct 2026

What Re-ran the whole QAOA sweep in CUDA-Q (CPU simulator `qpp-cpu`) to check it against Qiskit at the largest tiny sizes, plus a cross-check of CUDA-Q at Qiskit's trained angles.

Results Cross-check: all 18 rows agree to 5.4×10^{-7}. Independent training: p = 1 and p = 2 land within 10^{-4} of Qiskit's angles, with identical approximation ratios. p = 3 was cancelled at 3 of 6 cases (all matching). CUDA-Q on CPU is about 80× slower than Aer (17 s vs 0.2 s per evaluation at 21 qubits); its fast simulators need a GPU.

QAOA results and notebook section (PR #22)

7 Oct 2026

What `run_qaoa_benchmark.py` runs depths 1–3 on the six tiny rows and writes `results/qaoa_tiny.csv`; a new QAOA section in `notebooks/analysis.ipynb` with two figures. Also scored the angles trained on `tiny_a`, reused unchanged on the other rows (the transfer baseline).

Results Approximation ratio 0.80–0.82 at p = 1, 0.92–0.93 at p = 3. P(optimal) relative to random: 6.1–8.5 at p = 3 on the larger B = 1 rows, but only 1.6–2.1 on the B = 2 rows at every depth. Transferred angles lose at most 0.0023 in approximation ratio.

PR #22: penalty QAOA in Qiskit and CUDA-Q

6–7 Oct 2026

What `qaoa_common.py` (shared Hamiltonian, scoring, training), `qaoa_qiskit.py` (Aer), `qaoa_cudaq.py` (CUDA-Q kernel), `requirements-qaoa.txt`, a separate CI job. Both backends are tested against an independent numpy simulation; 17 reintroduced bugs were all caught.

Results On `tiny_a` B=1, p = 2: P(optimal) 0.333 vs 0.125 for random guessing; both backends train to the same angles. At B = 2, lower energy does not mean better cuts.

Research: GAT and TransferTrainer (no code)

6 Oct 2026

Findings IBM's `TransferTrainer` is a lookup (features → stored angles), not a learned model; it can serve NIP with an NIP feature set, a nearest-match rule and no angle averaging. The MIS GAT work classifies a graph into a cluster of donor graphs, then tries the nearest donors' angles; it handles only unweighted graphs at p = 2. Neither is implemented yet.

P0: merges, slide fix, proposal sentence

6 Oct 2026

What Merged PR #20 (`c08a5f7`) and PR #21 (`48c8581`, after resolving one conflict). Corrected slide 2 of the deck: max flow 22 (not 25), reduction 77% (not 80%), saved as `st_NIP_corrected.pptx`. Posted the corrected proposal results sentence on #18.

Repository review and priority list

6 Oct 2026

What Read the code, issues and PRs; ranked next steps (P0 unblock, P1 hard instances and baselines, P2 experiments, P3 blocked items).